Machine Learning MCQ Question Bank

Refined version: Exact repeated questions removed; original MCQ style and terminology retained as far as possible. Correct answers are provided at the end.

1. Machine learning is an application of _____.

· A. Block Chain

· B. Artificial Intelligence

· C. Both A and B

· D. None of these

2. Which of the following are the Applications of Machine learning ?

· A. Email Filtering

· B. Sentiment Analysis

· C. Face Recognition

· D. All of the above

3. Machine learning is a subset of which of the following fields?

· A. Data Learning

· B. Deep Learning

· C. Artificial Intelligence

· D. None of these

4. In which industry is machine learning commonly used for fraud detection and risk management?

· A. Agriculture

· B. Finance

· C. Entertainment

· D. Education

5. What is the primary goal of supervised learning?

· A. To discover patterns in unlabeled data

· B. To predict outcomes based on labeled input data

· C. To reduce the dimensionality of the data

· D. To cluster similar data points

6. Which of the following techniques is used to prevent overfitting?

· A. Increasing the learning rate

· B. Cross-validation

· C. Reducing the training dataset size

· D. Using a linear model for complex data

7. Which algorithm is primarily used for dimensionality reduction?

· A. Logistic Regression

· B. Decision Trees

· C. Principal Component Analysis (PCA)

· D. Support Vector Machines

8. If mean is less than mode, the distribution will be?

· A. Positively skewed

· B. Negatively skewed

· C. Symmetrical

· D. None of these

9. An observation that lies an abnormal distance from other values in a random sample from a population is refer to as?

· A. Syrical point

· B. External point

· C. Outlier

· D. Skew point

10. What is the primary purpose of EDA in Data Science?

· A. To make data more complex

· B. To simplify complex data

· C. To discover insights & patterns in data

· D. To visualize data

11. What is the purpose of encoding categorical data in machine learning?

· A. To increase the size of the dataset.

· B. To reduce the number of categories

· C. To make the data human-readable

· D. To represent categorical data in a format suitable for machine learning algorithms

12. How can outliers in a numerical dataset be treated?

· A. Ignoring them during analysis

· B. Replacing them with the median value

· C. Removing them or transforming them

· D. Assigning them a weight of 0

13. Which method in pandas provides a concise summary of a DataFrame, including the number of non-null values in each column?

· A. df.describe()

· B. df.info()

· C. df.head()

· D. df.shape()

14. Linear Regression is an example of?

· A. Supervised Learning

· B. Unsupervised Learning

· C. Semi-Supervised Learning

· D. Reinforcement Learning

15. For what Polynomial Regression is used?

· A. Find the best linear line

· B. Handle with non-linear and separable data

· C. Handle linear and separable data

· D. Classify binary data

16. What is the name of a regression model in which more than one independent variable is utilized to predict the dependent variable?

· A. a simple linear regression model

· B. a multiple regression model

· C. an independent model

· D. none of the above

17. What happens when we introduce more variables to a linear regression model?

· A. The r squared value may increase or remain constant, the adjusted r squared may increase or decrease

· B. The r squared may increase or decrease while the adjusted r squared always increases.

· C. Both r square and adjusted r square always increase on the introduction of new variables in the model.

· D. Both might increase or decrease depending on the variables introduced.

18. What does the assumption of Linearity in machine learning imply?

· A. The relationship between features and target variable is not linear

· B. The model should be a straight line

· C. The residuals should be normally distributed

· D. The data should be linearly separable

19. What is the primary purpose of a Loss Function in Linear Regression?

· A. To maximize the accuracy of predictions

· B. To minimize the difference between predicted and actual values

· C. To calculate the mean of the target variable

· D. To identify outliers in the data

20. Ridge and Lasso regression are simple techniques to _____the complexity of the model and prevent over-fitting which may result from simple linear regression.

· A. Increase

· B. Eliminate

· C. Decrease

· D. None of the above

21. Ridge regression uses which norm?

· A. L1

· B. L2

· C. Combination of L1 and L2

· D. None

22. Which one is true?

· A. Ridge and Lasso regression are techniques to reduce the model complexity and prevent over-fitting which may result from simple linear regression

· B. Ridge regression shrinks the coefficients and it helps to reduce the model complexity and multi-collinearity.

· C. Lasso regression not only helps in reducing over-fitting but it can help us in feature selection

· D. All of the above

23. Which of the following of the coefficients is added as the penalty term to the loss function in Lasso regression?

· A. Absolute value of magnitude

· B. Squared magnitude

· C. Number of non-zero entries

· D. None of the above

24. What type of penalty is used on regression weights in Ridge regression?

· A. L0

· B. L2

· C. L1

· D. None of the above

25. In Ridge regression, A hyper parameter is used called _____ that controls the weighting of the penalty to the loss function.

· A. Gamma

· B. Alpha

· C. Lambda

· D. None of above

26. In terms of the bias-variance trade-off, which of the following is substantially more harmful to the test error than the training error?

· A. Bias

· B. Loss

· C. Variance

· D. Risk

27. Which of the following is a common error measure?

· A. Median absolute deviation

· B. Sensitivity

· C. Specificity

· D. All of the mentioned

28. For Ridge Regression, if the regularization parameter = 0, what does it mean?

· A. Large coefficients are not penalized

· B. Overfitting problems are not accounted for

· C. The loss function is as same as the ordinary least square loss function

· D. All of the above

29. For Lasso Regression, if the regularization parameter = 0, what does it mean?

· A. The loss function is as same as the ordinary least square loss function

· B. Can be used to select important features of a dataset

· C. Shrinks the coefficients of less important features to exactly 0

· D. All of the above

30. What is the main purpose of Ridge and Lasso regularization in linear regression?

· A. To increase the complexity of the model

· B. To reduce the impact of outliers

· C. To penalize large coefficients and prevent overfitting

· D. To simplify the model by removing unnecessary features

31. What is the main purpose of cross-validation in machine learning?

· A. To increase the training time of the mode

· B. To assess how well the model will generalize to an independent dataset

· C. To maximize the number of features in the model

· D. To ensure the model perfectly fits the training data

32. Logistic regression assumes a:

· A. Linear relationship between continuous predictor variables and the outcome variable.

· B. Linear relationship between continuous predictor variables and the logit of the outcome variable.

· C. Linear relationship between continuous predictor variables.

· D. Linear relationship between observations.

33. divide the total number of correctly classified positive examples by the total number of predicted positive examples. Which is this formula?

· A. Precision

· B. Accuracy

· C. Recall

· D. None of these

34. Formula - TP / (TP + FN): Which of these is being represented by formula?

· A. Accuracy

· B. Recall

· C. harmonic mean

· D. Classification Rate

35._____uses Harmonic Mean in place of Arithmetic Mean as it punishes the extreme values more.

· A. S-measure

· B. F-measure

· C. T-measure

· D. None of these

36. What does the coefficient of a feature in logistic regression represent?

· A. The variance explained by that feature

· B. The change in log odds for a one-unit increase in the feature

· C. The probability of the target class

· D. The correlation between the feature and the target

37. What is the logistic loss function also known as?

· A. Mean Squared Error

· B. Hinge Loss

· C. Cross-Entropy Loss

· D. Euclidean Loss

38. What method is commonly used to estimate the coefficients in logistic regression?

· A. Least Squares

· B. Maximum Likelihood Estimation (MLE)

· C. k-means clustering

· D. Ridge Regression

39. What is the purpose of the ROC curve in the context of logistic regression?

· A. To measure feature importance

· B. To visualize the trade-off between sensitivity and specificity

· C. To determine the best-fit line

· D. To check for multicollinearity

40. Decision Tree is a display of an algorithm?

· A. True

· B. False

41. In Decision tree pruning methods include?

· A. Prunning

· B. Boosting

· C. Bagging

· D. All of the above

42. Decision tree builds regression or classification models in the form of a tree structure. It breaks down a dataset into smaller and smaller subsets while at the same time an associated decision tree is incrementally developed. The final result is a tree with_____and _____

· A. Tree nodes and leaf nodes

· B. branch nodes and leaf nodes

· C. decision nodes and leaf nodes

· D. All of these

43. Decision tree can be used for _____.

· A. classification

· B. regression

· C. Both

· D. None of these

44. In a decision tree, what does a leaf node represent?

· A. A decision point

· B. A split in the data

· C. An output class or value

· D. An attribute of the data

45. What is the primary purpose of the k-NN algorithm?

· A. Clustering data points

· B. Predicting continuous values

· C. Classifying data points based on similarity

· D. Reducing dimensionality

46. How does k-NN determine the "nearest" neighbors?

· A. By calculating the average distance

· B. By using a fixed distance metric

· C. By randomly selecting points

· D. By performing a clustering analysis

47. What happens if the value of k is set too low in k-NN?

· A. The model becomes too generalized.

· B. The model may overfit the training data.

· C. The computational cost increases significantly.

· D. The model cannot make predictions.

48. Which of the following is a drawback of the k-NN algorithm?

· A. It is easy to implement.

· B. It can handle large datasets efficiently.

· C. It cannot handle large datasets efficiently.

· D. It requires extensive training time.

49. In k-NN, how is the final prediction for a new data point made?

· A. By averaging the features of the k nearest neighbors

· B. By taking the median of the target values of the k nearest neighbors

· C. By voting among the classes of the k nearest neighbors

· D. By using a weighted average of all data points

50. What is a common method to determine the optimal value of k in k-NN?

· A. Using the mean of the dataset

· B. Cross-validation

· C. Random selection

· D. Heuristic guessing

51. What is the effect of using a very small value of k (e.g., k=1)?

· A. The model becomes more generalized.

· B. The model is likely to be more sensitive to noise.

· C. The model will perform equally well on training and test data.

· D. The model's predictions will be very stable.

52. What do you mean by a hard margin?

· A. The SVM allows very low error in classification

· B. The SVM allows high amount of error in classification

· C. The SVM allows no error in classification

· D. none of these

53. Support vectors are the data points that lie closest to the decision surface.

· A. False

· B. True

54. Support Vector Machine(SVM) can be used for both classification or regression challenges. Which kind of learning technique SVM uses?

· A. unsupervised

· B. supervised

· C. reinforced

· D. clustered

55. which approach breaks down the multiclass problem into multiple binary classification problems?

· A. Two-to-One approach

· B. One-to-One approach

· C. One-to-Many approach

· D. One-to- two approach

56. _____ calculates the harmonic mean between precision and recall, and both depend on the false positive and false negative

· A. f1 score

· B. f2 score

· C. z score

· D. z1 score

57. SVM stands for _____?

· A. support vector machine

· B. support vector mapping

· C. special vector machine

· D. special vector mentor

58. In its most simple type, SVM doesn't support multiclass classification natively.

· A. TRUE

· B. FALSE

59. In Which approach, the breakdown is set to a binary classifier per each class?

· A. One-to-Rest

· B. One-to-Many

· C. One-to-One approach

· D. None of these

60. Does gradient boosted trees generally perform better than random forest?

· A. Yes

· B. No

· C. Can't Say

61. Which regularization technique is used to penalize complex models?

· A. L1 regularization

· B. L2 regularization

· C. Both A and B

· D. None of these

62. This feature optimizes the available memory space and maximizes its usage when handling huge datasets that don't fit into memory. Which feature is this?

· A. Regularization

· B. Out-of-core computing:

· C. Handling sparse data

· D. All of these

63. Adaboost selects a training subset randomly?

· A. TRUE

· B. FALSE

64. _____ relies on the intuition that the best possible next model, when combined with previous models, minimizes the overall prediction error

· A. Gradient Boost

· B. XG Boost

· C. Ada Boost

· D. All of these

65. What is the primary purpose of Principal Component Analysis (PCA)?

· A. To increase the dimensionality of the data

· B. To visualize high-dimensional data in lower dimensions

· C. To perform clustering

· D. to classify data points

66. What does PCA do to the original features in the data?

· A. It creates new features that are linear combinations of the original features.

· B. It selects a subset of the original features.

· C. It transforms the features into categorical values.

· D. It normalizes the features to a common scale.

67. What is a common preprocessing step before applying PCA?

· A. Data augmentation

· B. Feature selection

· C. Normalization or standardization

· D. Clustering

68. What is the main goal of dimensionality reduction?

· A. To increase the number of features

· B. To reduce the complexity of data while preserving its essential characteristics

· C. To eliminate all features

· D. To classify data points

69. Which of the following is a common method for dimensionality reduction?

· A. Decision Trees

· B. Support Vector Machines

· C. Principal Component Analysis (PCA)

· D. k-Nearest Neighbors

70. Which of the following is NOT a common technique for dimensionality reduction?

· A. Linear Discriminant Analysis (LDA)

· B. t-Distributed Stochastic Neighbor Embedding (t-SNE)

· C. Gradient Descent

· D. Autoencoders

71. What do the eigenvalues represent in PCA?

· A. The number of features in the dataset

· B. The variance explained by each principal component

· C. The distances between data points

· D. The direction of the data spread

72. What is the primary goal of K-means clustering?

· A. To classify data points into pre-defined categories

· B. To minimize the sum of squared distances between data points and their assigned cluster centers

· C. To maximize the distance between different clusters

· D. To identify outliers in the dataset

73. How does K-means initialize the centroids?

· A. Randomly selecting K data points as initial centroids

· B. Choosing the data point closest to the mean

· C. Using the first K data points in the dataset

· D. Setting all centroids to zero

74. What is the main purpose of the Elbow Method in K-means clustering?

· A. To visualize the clusters

· B. To determine the optimal number of clusters (K)

· C. To compute the centroids

· D. To evaluate the clustering performance

75. What type of plot is typically used in the Elbow Method?

· A. Scatter plot

· B. Bar chart

· C. Line plot

· D. Box plot

76. Which distance measure is most commonly used in K-means clustering?

· A. Manhattan Distance

· B. Chebyshev Distance

· C. Euclidean Distance

· D. Cosine Similarity

77. What is a characteristic of Manhattan Distance compared to Euclidean Distance?

· A. It always yields a larger value.

· B. It always yields a larger value.

· C. It sums the absolute differences of coordinates.

· D. It is always less sensitive to outliers.

78. Which distance measure would be most appropriate for categorical data?

· A. Euclidean Distance

· B. Hamming Distance

· C. Manhattan Distance

· D. Minkowski Distance

79. In agglomerative hierarchical clustering, what is the initial step?

· A. Choosing the number of clusters

· B. Calculating the distance matrix

· C. Merging the closest clusters

· D. Assigning data points to their nearest cluster

80. What is one disadvantage of hierarchical clustering?

· A. It is sensitive to noise and outliers.

· B. It cannot visualize clusters.

· C. It does not provide a hierarchical structure.

· D. It is computationally inexpensive.

81. Which of the following best describes agglomerative hierarchical clustering?

· A. It starts with a single cluster containing all data points.

· B. It starts with each data point as its own cluster and merges them iteratively.

· C. It splits a large cluster into smaller clusters.

· D. It requires the number of clusters to be defined before clustering.

82. In divisive hierarchical clustering, what is the initial step?

· A. Merging the closest clusters

· B. Creating a distance matrix

· C. Starting with a single cluster containing all data points

· D. Assigning data points to clusters

83. Which of the following is a key advantage of divisive hierarchical clustering over agglomerative clustering?

· A. It provides more detailed cluster hierarchies.

· B. It is faster for large datasets.

· C. It is less sensitive to outliers.

· D. It can adaptively determine the number of clusters.

84. Which of the following describes complete linkage in hierarchical clustering?

· A. It averages the distances between all points in two clusters.

· B. It uses the minimum distance between points in the clusters.

· C. It measures the maximum distance between points in the two clusters.

· D. It calculates the centroid of each cluster.

85. What is the main goal of cross-validation?

· A. Increase training time

· B. Reduce the size of dataset

· C. Estimate the performance of a model

· D. Improve the testing dataset

86. In Leave-One-Out Cross Validation (LOOCV), the number of folds equals:

· A. Total rows in dataset

· B. 10

· C. Number of features

· D. Number of models

87. What is the purpose of hyperparameter tuning?

· A. Change training data

· B. Improve model evaluation metric

· C. Add more features

· D. Train on test set

88. What does GridSearchCV do?

· A. Randomly selects hyperparameters

· B. Selects best model automatically

· C. Tests all combinations of given hyperparameters

· D. Applies unsupervised learning

89. Compared to GridSearchCV, RandomizedSearchCV is:

· A. Slower

· B. More accurate always

· C. Faster and tests fewer combinations

· D. Used for classification only

90. What is returned by GridSearchCV after fitting?

· A. Only accuracy

· B. Best model and parameters

· C. Confusion matrix

· D. Only test results

91. High bias leads to

· A. Overfitting

· B. Underfitting

· C. Optimal generalization

· D. Large training dataset

92. What does the bias-variance tradeoff aim to balance?

· A. Dataset and algorithm

· B. Overfitting and underfitting

· C. Model speed and size

· D. Accuracy and runtime

93. Which model is most likely to have high variance?

· A. Linear Regression

· B. k-NN with k=1

· C. Decision Tree with max_depth=1

· D. Logistic Regression

94. What happens when a model is too complex?

· A. High bias

· B. Underfitting

· C. High variance

· D. No change

95. Which programming language is primarily used with Streamlit?

· A. Java

· B. Python

· C. R

· D. C++

96. How do you run a Streamlit app?

· A. python run app.py

· B. streamlit run app.py

· C. python app.py

· D. streamlit start app.py

97. What file format does a Streamlit script use?

· A. .doc

· B. .py

· C. .exe

· D. .txt

98. Can you visualize charts in Streamlit?

· A. No

· B. Yes

· C. Only images

· D. Not supported

99. What is the purpose of 'st.slider()' in Streamlit?

· A. Display a button

· B. Input numerical values

· C. Show messages

· D. Create dropdown

100. What is the first step in deploying a machine learning model as a web app?

· A. Training the model

· B. Saving model with joblib or pickle

· C. Cleaning data

· D. Create Streamlit file

101. What is the main objective of Machine Learning?

· A. Creating intelligent machines with emotions

· B. Writing programs to perform specific tasks.

· C. Teaching computers to learn from data and improve performance

· D. Designing hardware for computational tasks.

102. What does overfitting refer to in machine learning?

· A. A model that performs well on both training and test data

· B. A model that learns noise from the training data instead of the underlying pattern

· C. A model that is too simple to capture the data's complexity

· D. A model that generalizes well to unseen data

103. If mean, median, and mode are all equal then distribution will be?

· A. Negative Skewed

· B. Symmetrical

· C. Circular

· D. Positive Skewed

104. Which of the following is not a major data analysis approaches?

· A. Data Mining

· B. Business Intelligence

· C. Predictive Intelligence

· D. Text Analytics

105. Which of the following information is not given by Five Number Summary?

· A. Minimum

· B. Median

· C. Mode

· D. All of the above

106. The term "Correlation" in EDA refers to ?

· A. Measure of central tendency

· B. Measure of spread of data

· C. Measure of relationship between variables

· D. Measure of data distribution

107. _____is an analytical approach to Linear Regression with a Least Square Cost Function..

· A. Slope Equation

· B. Variable equation

· C. Normal Equation

· D. none of these

108. Which of the following is a commonly used metric for evaluating the performance of a Linear Regression model?

· A. Accuracy Score

· B. F1 Score

· C. Mean Squared Error (MSE)

· D. Confusion Matrix

109. What is a key assumption of Linear Regression?

· A. The data must have a normal distribution

· B. The relationship between variables is linear

· C. Outliers have no impact on the model

· D. The number of features should be equal to the number of observations

110. What's the penalty term for the Ridge regression?

· A. the square of the magnitude of the coefficients

· B. the square root of the magnitude of the coefficients

· C. the absolute sum of the coefficients

· D. the sum of the coefficients

111. With Lasso Regression the influence of the hyper parameter lambda, as lambda tends to zero the solution approaches to

· A. Zero

· B. One

· C. Linear Regression

· D. Infinity

112. What's the cost function of the logistic regression?

· A. Sigmoid function

· B. Logistic Function

· C. both (A) and (B)

· D. none of these

113. What is the purpose of performing cross-validation?

· A. To assess the predictive performance of the models

· B. To judge how the trained model performs outside the sample on test data

· C. Both A and B

· D. None of these

114. What is the primary purpose of logistic regression in machine learning?

· A. To perform clustering

· B. To predict continuous values

· C. To model binary outcomes

· D. To reduce dimensionality

115. Which function does logistic regression use to model the probability of the target class?

· A. Linear function

· B. Step function

· C. Sigmoid function

· D. Polynomial function

116. What is the main purpose of a decision tree in machine learning?

· A. To perform clustering

· B. To reduce dimensionality

· C. To make predictions based on input features

· D. To visualize data distributions

117. What is a common criterion used to split nodes in a decision tree for classification tasks?

· A. Mean Squared Error

· B. Gini Impurity

· C. Kullback-Leibler Divergence

· D. Correlation Coefficient

118. What is a potential drawback of decision trees?

· A. They can handle both categorical and numerical data.

· B. They are easy to interpret and visualize.

· C. They can easily overfit the training data.

· D. They require less computational power than other algorithms.

119. How does increasing the value of k affect the decision boundary in k-NN?

· A. It makes the decision boundary more complex.

· B. It makes the decision boundary smoother.

· C. It has no effect on the decision boundary.

· D. It always makes the decision boundary linear.

120. In SVM, what is a hyperplane?

· A. decision boundaries

· B. data points

· C. features

· D. none of these

121. SVM is a supervised Machine Learning can be used for_____?

· A. Classification

· B. Regression

· C. Either a or b

· D. None of these

122. Closest Point to the hyper plane are support vectors?

· A. False

· B. True

· C. Unpredictable

· D. None of these

123. What is the primary goal of SVM in Machine Learning?

· A. To maximize the margin between classes

· B. To minimize the number of support vectors

· C. To Visualize data relationships

· D. To perform Unsupervised Learning

124. Below lists some heuristics for best preparing your data for AdaBoost. Which of the following is an approach?

· A. Quality Data

· B. Outlier

· C. Noisy Data

· D. All of these

125. Unlike in AdaBoost, the incorrect result is given a higher weightage in gradient boosting.

· A. TRUE

· B. FALSE

126. Why is dimensionality reduction important in machine learning?

· A. It guarantees higher accuracy for all algorithms.

· B. It reduces computational cost and mitigates the curse of dimensionality.

· C. It simplifies the model structure without losing any information.

· D. It can increase the number of features available for training.

127. What is the "curse of dimensionality"?

· A. The phenomenon where data becomes less complex as dimensionality increases

· B. The challenge that arises from analyzing and organizing data in high dimensions, leading to sparse data and overfitting

· C. The ability of models to perform better with more features

· D. The reduction of computational requirements with increased dimensions

128. How is the covariance matrix calculated in PCA?

· A. By summing all feature values

· B. By averaging the product of feature pairs

· C. By finding the correlation coefficients between features

· D. By multiplying the feature matrix by its transpose

129. What role do eigenvectors play in PCA?

· A. They define the scale of the data.

· B. They determine the importance of features.

· C. They represent the directions of maximum variance in the data.

· D. They normalize the data.

130. After calculating the eigenvalues and eigenvectors, what is the next step in PCA?

· A. Project the original data onto the eigenvectors.

· B. Select the features based on their eigenvalues.

· C. Normalize the eigenvectors.

· D. Calculate the mean of the original features.

131. Which of the following is a necessary input for K-means clustering?

· A. The number of clusters (K)

· B. The feature names

· C. The data labels

· D. The distance metric

132. Which of the following is a limitation of K-means clustering?

· A. It can handle large datasets efficiently

· B. It assumes spherical clusters

· C. It works well with any distance metric

· D. It is easy to implement

133. If the WCSS continues to decrease without an elbow in the plot, what might this indicate?

· A. The dataset is well-clustered

· B. The optimal K is very high

· C. The clusters are too small

· D. There are no meaningful clusters

134. Which of the following is a limitation of the Elbow Method?

· A. It requires a large amount of data

· B. The elbow point can be subjective

· C. It can only be applied to spherical clusters

· D. It provides exact values for K

135. Which distance measure would be more suitable when dealing with high-dimensional sparse data, such as text data represented by TF-IDF vectors?

· A. Euclidean Distance

· B. Manhattan Distance

· C. Cosine Similarity

· D. Jaccard Distance

136. What is the main characteristic of hierarchical clustering?

· A. It requires the number of clusters to be specified in advance.

· B. It builds a hierarchy of clusters either in a bottom-up or top-down manner.

· C. It is primarily used for large datasets.

· D. It only uses Euclidean distance for distance measurement.

137. What is a dendrogram in the context of hierarchical clustering?

· A. A statistical model to estimate cluster density

· B. A tree-like diagram that illustrates the arrangement of clusters

· C. A distance metric used to measure similarity

· D. A method for dimensionality reduction

138. Which of the following statements about agglomerative and divisive clustering is true?

· A. Both methods require the number of clusters to be specified in advance.

· B. Agglomerative clustering is more computationally intensive than divisive clustering.

· C. Divisive clustering typically requires more computational resources than agglomerative clustering.

· D. Both methods always yield the same clustering results.

139. What is the primary criterion for merging clusters in agglomerative clustering?

· A. The average distance between all points in the clusters

· B. The maximum distance between points in the clusters

· C. The minimum distance between points in the clusters

· D. The distance between the closest pair of points in each cluster

140. What does "single linkage" refer to in hierarchical clustering?

· A. The distance between the farthest points of two clusters.

· B. The distance between the closest points of two clusters.

· C. The average distance between all points in two clusters.

· D. The variance within a cluster.

141. What is the main characteristic of average linkage in hierarchical clustering?

· A. It uses the closest points to define the distance between clusters.

· B. It averages the distances between all points in the two clusters.

· C. It only considers the centroid of each cluster.

· D. It is not commonly used in practice.

142. Which linkage method tends to create elongated clusters and is sensitive to noise?

· A. Complete linkage

· B. Average linkage

· C. Single linkage

· D. Ward's linkage

143. What is Ward's linkage method primarily designed to achieve in hierarchical clustering?

· A. Minimize the distance between the closest points in clusters.

· B. Minimize the total within-cluster variance.

· C. Maximize the distance between clusters.

· D. Create equal-sized clusters.

144. In k-fold cross-validation, the data is divided into:

· A. 2 parts

· B. 3 parts

· C. k equal parts

· D. Random number of parts

145. Which of the following is a limitation of simple train/test split?

· A. It is easy to implement

· B. It gives high accuracy

· C. Model may overfit

· D. Results can vary depending on split

146. Which parameter of GridSearchCV defines how many folds to use in CV?

· A. param_grid

· B. cv

· C. scoring

· D. verbose

147. What is Streamlit used for?

· A. Data collection

· B. Database creation

· C. API development

· D. None of the above

148. Which function is used to display text in Streamlit?

· A. print()

· B. st.write()

· C. show()

· D. display()

149. Which port does Streamlit usually run on by default?

· A. 3000

· B. 5000

· C. 8501

· D. 8888

150. What is Machine learning?

· A. The autonomous acquisition of knowledge through the use of computer program

· B. The autonomous acquisition of knowledge through the use of manual programs

· C. The selective acquisition of knowledge through the use of computer programs

· D. The selective acquisition of knowledge through the use of manual programs

151. Which of the following is a common metric for evaluating classification models?

· A. Mean Squared Error

· B. R-squared

· C. Accuracy

· D. Silhouette Score

152. In EDA, which Measure of central tendency is typically represented by the height of a box in a Box Plot?

· A. Mean

· B. Median

· C. Mode

· D. Variance

153. How many coefficients do you need to estimate in a simple linear regression model (One independent variable)?

· A. 1

· B. 12

· C. 3

· D. 2

154. True-False: Linear Regression is mainly used for Regression.

· A. TRUE

· B. FALSE

155. _____loss is mostly used in SVM, this is used in the combination of the activation function in the last layer. We use this loss to classify whether an email is a spam or not.

· A. Hinge loss

· B. entropy loss

· C. cross-entropy loss

· D. MAE loss

156. Which of the following is correct use of cross validation?

· A. Selecting variables to include in a model

· B. Selecting parameters in prediction function

· C. Comparing predictors

· D. All of these

157. Elastic Net uses which Norm?

· A. L1

· B. L2

· C. Both L1 & L2

· D. None of the above

158. When should you consider using a larger value of k?

· A. When the dataset is small

· B. When the dataset contains a lot of noise

· C. When the classes are well-separated

· D. When computational efficiency is not a concern

159. What is the potential downside of using a very large value of k in k-NN?

· A. Increased risk of overfitting

· B. Increased model interpretability

· C. Loss of important local patterns

· D. Decreased computational cost

160. Which of the following is/are true about boosting trees?
1. In boosting trees, individual weak learners are independent of each other.
2. Boosting improves performance by combining weak learners.

· A. 2

· B. 1

· C. 1 & 2

· D. 1^2

161. _____is an implementation of gradient boosted decision trees designed for speed and performance.

· A. Ada Boost

· B. Gradient Boost

· C. XGBoost

· D. All of these

162. in the case of AdaBoost, very short decision trees were used that only had a single split, called a _____?

· A. optimizer step

· B. Decision stump

· C. weak learner

· D. bench step

163. What does the proportion of variance explained by each principal component indicate?

· A. The importance of the feature in the original dataset

· B. The contribution of that component to the overall variance in the data

· C. The degree of correlation between features

· D. The potential for overfitting

164. In the Elbow Method, what does the "elbow" point indicate?

· A. The best K value with the lowest WCSS

· B. The point where increasing K does not significantly reduce WCSS

· C. The maximum number of clusters

· D. The first point in the WCSS plot

165. Which linkage method considers the maximum distance between points in two clusters?

· A. Single linkage

· B. Complete linkage

· C. Average linkage

· D. Ward’s linkage